

Applied data Analytics approach for defect root causes analysis in manufacturing: the case of multi-product assembly lines

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Abstract—Multi-product assembly lines are commonly used for their flexibility in mass-customized production, but their complexity makes identifying the root causes of defects difficult. This research addresses the limitations of traditional methods for analyzing the root causes of defects in this context. It introduces a four-step methodology for preparing data and conducting descriptive analysis of defect root causes. Product families are created by segmenting production data from the company's information systems using AHC algorithms or company knowledge. Defect rates are then calculated for each product group and the transitions between product sequences. Finally, a CART decision tree is used to generate rules that lead to defective clusters. These rules are used for descriptive analysis of defect root causes and are seen as improvement opportunities for multi-product assembly lines. The methodology was applied to two case studies using real production data. This led to the identification and validation of the root causes of defects by the partner companies. Nevertheless, limitations must be taken into account, e.g., the reliance on expert judgment for the identification of root causes, the sensitivity of the data used, and the necessity of regenerating the decision tree for each new analysis.

Index Terms—multi-product assembly line, root cause analysis, defect, machine learning, manufacturing

I. INTRODUCTION

To meet the expectations of a demanding, volatile, and competitive market, today's manufacturers face significant challenges in achieving cost efficiency and mass customization [1, 2]. To meet the need for mass customization, mixed-model assembly lines have been developed [3]. They can produce a large number of part numbers, but their complexity can be detrimental to production quality due to the overloading of workstations [3]. Yet production quality is one of the most important factors in cost efficiency in this context.

For instance, in the automotive industry in 2004, over 1,000€ were spent per vehicle to manage defective parts [4]. It is believed that this figure has increased since then [5]. Thus, identifying the root cause of defects is crucial in manufacturing

since it may reduce production risks and unexpected downtime impacts, thus optimizing the production process [6].

As Industry 4.0 develops and data becomes more prevalent in production [7], new data enrichment techniques are emerging to detect defects [8, 9, 10, 11] and their root causes [12, 13, 14, 15]. Most of machine learning and artificial intelligence approaches rely on models that require several examples of an event occurrence to be trained. However, the high variability of products in contemporary production lines considerably reduces the number of defects per reference present in the databases, thus generating highly unbalanced datasets. Papageorgiou et al. [6] note that while there is a significant amount of research on manufacturing issues, little attention has been paid to assembly lines, specifically in identifying the root causes of defects on multi-product assembly lines.

To fill this gap, the present paper proposes a methodology developed using real data retrieved from multiple machines and production cells of the automotive industry. It leverages data analytics to improve the identification of root causes of defects on multi-product assembly lines. The aim is not the detection of defective vehicles but rather the use of the power of data to help automate a part of the complex process of identifying the most probable combination of factors that can lead to defects on the assembly lines. This research paves the way for data-driven identification of the root causes of defects on multi-product assembly lines. It facilitates the identification of manufacturing process improvement opportunities, thus combining cost-effectiveness and mass customization.

A brief overview of multi-product assembly line concepts and defect root cause analysis is presented in section II. The proposed methodology is explained in section III while the results obtained from its application on case studies are presented in section IV. These results are discussed in section V and section VI concludes the research.

II. STATE OF THE ART

A. Multi-product assembly lines

Assembly lines are an interesting way of achieving low-cost mass production [16]. Traditional assembly lines specialize in the mass production of a single product, which may not align with the short product life cycles and high product variety demands present in some market environments. In such cases, a large number of different models have to be produced in relatively small batches [17].

The multi-product assembly line, also known as a mixed-model assembly line, represents a dynamic paradigm in manufacturing that accommodates the production of multiple product variants on a single assembly line [18]. Such production lines make it possible to produce diversified small batches and respond rapidly to changes in demand for a variety of models while maintaining low inventories [19]. To do so, a variety of tasks must be performed on the line to produce different variants that differ in specific characteristics or technical requirements [16].

The ability of the multi-product assembly line to seamlessly switch between different product configurations while maintaining high productivity and minimizing downtime is particularly valuable in industries characterized by varying customer demands and a need for customization [20].

Applications of the multi-product assembly can be found in automotive manufacturing [21, 20] and consumer goods [20] such as furniture manufacturing [22].

B. Classical defect root cause analysis

Root Cause Analysis (RCA) is a critical manufacturing process that helps identify the true cause of a problem, which is necessary to improve manufacturing operations [23].

Andersen [24] describes a general causal analysis process consisting of five successive steps. The problem is first identified by observing symptoms (1), then (2) the available information is collected to describe the problem in question. This is followed by (3) the identification of all possible causes leading to the problem. Finally, thanks to the knowledge of domain experts, (4) the root cause of the problem is identified, and (5) solutions (reactive or proactive) can be proposed and implemented.

This general analysis framework, aided by conventional tools such as the *Ishikawa diagram*, the *Failure Mode and Effect Analysis (FMEA)*, or the *5-why analysis*, is widely used in industry [25]. However, despite these tools, RCA analysis is a time-consuming and labor-intensive task [26]. Therefore, the use of such tools has a significant cost and is not without limitations. For instance, these tools are designed to focus on a particular process or product and may take time to produce convincing results [13]. They also rely heavily on experts, and root causes are often identified subjectively, without recourse to actual events recorded in the data [25].

C. Data-Driven root cause analysis

The rise of automation and Industry 4.0 has led to an exponential increase in available data, which presents both

challenges and opportunities for automating root cause analysis [23]. Data mining and machine learning tools can be used to make this process more efficient and help analysts identify the root cause of a problem [6, 23].

Convolutional neural networks (CNN) and decision trees are the most widely used tools for data-driven defect root causes analysis [6]. While CNN is widely used to detect defects by analyzing images such as in [10, 9], Decision trees (DTs) are widely used to link factors to observed problems because of their interpretative structures [23].

For example, the C5.0 decision tree algorithm is used to analyze anomalies in the roughness of a borehole and associate them with the process parameters responsible [27]. DTs are preferred to ensemble models such as Random Forest (RF) and Gradient Boosted Trees models because they offer a better compromise between performance and interpretability for identifying the defect root cause on low-capacity cell phone batteries [11]. These studies are based on all the products present in the production data to determine the root causes of defects. Other studies used decision trees coupled with clustering algorithms to group products or sequences of events to increase the quantity of data to be analyzed. For instance, a decision tree is associated with the hierarchical clustering algorithm (AHC) to promptly detect and report anomalies on an electromagnetic actuator assembly line [13].

In the context of assembly lines, the application of fault detection methods based on machine learning faces two main obstacles [28]:

- 1) The volume of data and variety of forms to process
- 2) Defective parts are difficult to identify because they are underrepresented in the database (low defect rate).

These difficulties are all the more acute in the case of multi-product assembly lines, where production variability and defect causation factors are all the greater. To the best of our knowledge, no scientific paper has proposed an algorithm for identifying the root cause of defects in the context of multi-product assembly lines. Yet this type of assembly line is one of the solutions adopted by manufacturers to meet the need for mass customization.

Automating root cause analysis in the manufacturing environment requires expert validation of the root causes identified by data-driven methods in the majority of research studies [23]. Therefore, these methods should be considered as providing potential root causes for experts to consider.

Considering the literature, there is a lack of research on root cause analysis methods for multi-product assembly lines. This paper proposes a root cause analysis method specific to this context to address this deficiency.

III. METHODOLOGY

A four-step methodology, illustrated in Figure 1, is proposed to identify the root causes of defects on multi-product assembly lines.

For the methodology to be applied, data regarding product specifications (*in blue*), and the production quality (*in green*) as well as the scheduling data (*in yellow*) must be available.

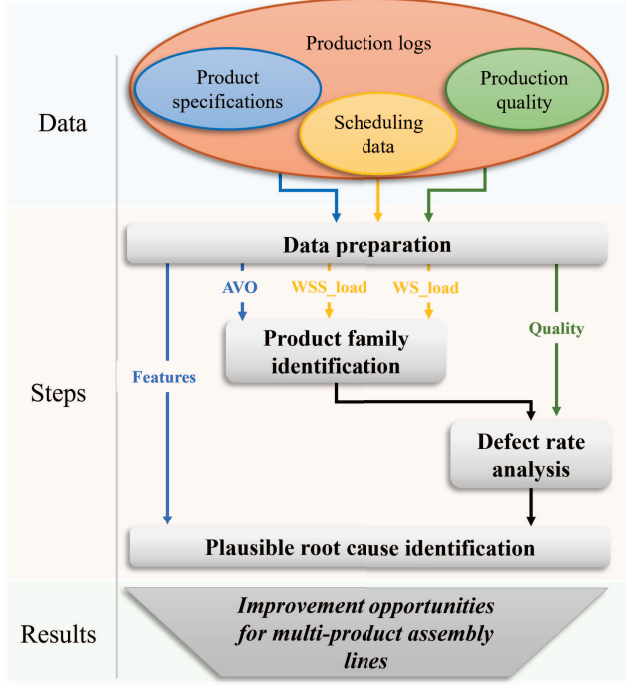


Fig. 1. Four-step methodology for root-cause defect identification on multi-product assembly lines

Usually it can be extracted from the information systems of the manufacturer or derived from production logs (*in red*).

A. Data preparation

In this section, extraction and feature engineering of pertinent information extracted from available data sources are presented.

Information available in manufacturing can vary. However, products are usually characterized either by categorical attributes describing the principal characteristics of the products or by the numerical attributes recording the production parameters values used to assemble the products. When working with categorical attributes, binarization of attributes to generate a **Features** dataset of size $[Number\ of\ unique\ products \times Number\ of\ unique\ specifications]$ is used.

Binary attributes describing whether or not each available operation has been performed on the products are also generated to form a dataset which will be defined as **AVO**. The characterization of products can also be completed by providing information regarding the operations performed on each assembled product. A new feature representing the normalized (Min/Max) number of operations performed on each assembled product is added to the **AVO** dataset. Thus, the size of the **AVO** dataset is $[Number\ of\ unique\ products \times Number\ of\ unique\ operations + 1]$.

A **WS_load** dataset providing information related to the workload needed to assemble each product per workstation is generated using available scheduling data. The workload

per workstation is calculated by combining the scheduling data with the **AVO** dataset, which specifies the operations performed on each assembled product, they are normalized using a Min/Max normalization. To determine what is considered to be an elementary operation, expert knowledge is required. Making this feature engineering task, a manual one. The size of the **WS_load** dataset is $[Number\ of\ unique\ products \times Number\ of\ workstations]$.

Depending on the organizational set-up, the workload can be defined at various organizational unit such as cost zones, assembly cells, etc. Being able to account for that is also relevant. Thus, another dataset named **WSS_load** corresponding to the sum of normalized workloads of workstations s belonging to the same organizational unit SS can be generated. The equation 1 is used to calculate the **WSS_load** from the **WS_load**.

$$WSS_load = \sum_{s \in SS} WS_load_s \quad (1)$$

Where :

WS_load_s = Normalized workload at s

s = Workstation on the assembly line

SS = Organizational unit grouping a subset of workstations on the assembly line

The size of this dataset will be of $[Number\ of\ unique\ products \times Number\ of\ organizational\ unit]$ and can be used in the **Product family identification** step, as the **WS_load** dataset.

A last dataset regarding defects encountered during assembly should also be generated. This data set is defined as the **Quality** dataset of size $[Number\ of\ unique\ products \times 1]$. We consider that a product is defective as soon as one defect is recorded. Therefore, the dataset **Quality** is a binary dataset in which each product is described as defective or not. This data can be retrieved from quality control data or from the production logs that record events that may have been related to defects such as delays in the assembly process. Usually, production logs include defect causes that should be accounted for when analyzing the logs.

B. Product family identification

In order to be able to leverage a high variability of products on production lines and deal with a considerably low number of defects per reference present in the databases, a **Product family identification** step is proposed. This step aims to form groups of products with similar assembly processes, to gather examples of combinations of defect factors, and thus overcome the curse of dimensionality associated with the multi-product assembly line.

A limited and restricted number of product feature combinations can be used when such a set exists within the

company's knowledge. If not, the AHC algorithm is used to group products with similar operations (**AVO** dataset) or similar workloads (**WS_load** and **WSS_load** datasets).

To generate clusters of similar products, similarity between products is compiled using a distance matrix between products. The distance is calculated using the Euclidean distance between the vector of features representing the product. Clusters are formed in an ascending manner to minimize the sum of squared distance differences between products in each group. The number of clusters is determined by the silhouette method [29].

C. Defect rate analysis

Once the clusters have been generated, the defect rate of the products in each one of them is calculated in step **Defect rate analysis**. Clusters generated are linked to the **Quality** dataset to calculate two metrics.

The first metric, provided by equation 2, corresponds to the percentage of defective products in each cluster i .

$$Defect_i = \frac{VD_i}{V_i} \quad (2)$$

Where :

$Defect_i$ = Defect rate in cluster i

VD_i = Number of defective products in cluster i

V_i = Number of products in cluster i

This defect rate is compared with the average defect rate of the data set to identify problematic clusters (i.e., clusters with a particularly high defect rate).

The second metric extends the analysis to the manufacturing sequence. It is provided by equation 3.

$$Defect_{A \rightarrow B} = \frac{VD_{A \rightarrow B}}{V_{A \rightarrow B}} \quad (3)$$

Where :

$Defect_{A \rightarrow B}$ = Defect rate in transition from cluster A to cluster B

$VD_{A \rightarrow B}$ = Number of defective products from cluster B preceded by a product from cluster A

$V_{A \rightarrow B}$ = Number of products from cluster B preceded by a product from cluster A

For each pair of product clusters A-B created in the **Product family identification** step, the rate of defective products from cluster B assembled after a product from cluster A is computed.

Only clusters and transitions between clusters with a minimum of k occurrences in the production sequence are considered to ensure the robustness of the analyses. Thus, these analyses enable the identification of problematic product clusters

and transitions between clusters associated with high defect rates.

D. Plausible root cause identification

As the problematic clusters have been identified, the **Plausible root cause identification** can be identified.

The CART algorithm is used to separate problematic clusters from others using the **Features** dataset. The CART algorithm is chosen for its interpretability, classification performance, and computation speed.

The separation is based on the *Gini diversity index*, which measures how often a random element in the set would be misclassified if its label were randomly chosen according to the distribution of labels in the subset. The minimum impurity reduction imp is used as a stopping criterion for tree branch separation. The nodes of the decision tree identify the product specifications (production parameters or product characteristics) that govern the appearance of problem clusters.

Each cluster is defined by a set of logical rules based on the **Features** attributes that govern the separation of products in the tree. Analyzing the nodes that lead to a problematic cluster can help identify the root causes of defects in a multi-product assembly line. The highly complex context of manufacturing makes it difficult to perform RCAs, multi-product assembly lines can be extremely challenging for experts due to the high number of options to consider. Using this method highly reduces the plausible causes, leading to a more focused and time-efficient method. Making the RCA process more accessible thus leading to greater opportunities for improvement in the context of multi-product assembly lines.

IV. CASE STUDY

The proposed methodology was applied to two distinct case studies based on real production data to identify the root causes of defects on multi-product assembly lines of pneumatics and automobiles, respectively.

A. Root causes of defects in pneumatic production

Data from a set of t tires is selected and information associated with the same product references are grouped into analysis partitions. These partitions are examined independently throughout the remainder of the case study.

1) *Data preparation*: Production logs data are prepared. For each assembled tire, p attributes representing the value of the production parameters are gathered to generate the **Features** dataset of size $[t \times p]$.

Production logs data are also used to extract the quality information. During the study period, d of delays were recorded in c different categories. The data is filtered to consider only delays related to tire production quality, reducing the number of delays and categories to consider (we will call them $d8$ and $c*$). We assume that a product is considered defective if at least one delay related to the tire's production quality has occurred during production. Thus, at the end of the preparation steps, a binary data set **Quality** of size $[t \times 1]$ is generated to determine whether a tire is defective or not.

In this case study, the product family identification is based on a limited and restricted number of production parameter combinations known by the operators of the company. Hence the uselessness and non-generation of **AVO** and **WS_load** datasets in this particular case study.

2) *Product family identification*: As mentioned before, the products from the same partition are clustered based on the combinations of production parameters used during assembly.

This is achievable due to the relatively small number of production parameter combinations for each case. For instance, the production of all products within the same reference A (partition A) resulted in 12 unique combinations of 17 necessary production parameters, as represented in Figure 2.

490	181	451	460	494	16	495	492	493	491	452	457	488	487	454	453	489	Nombre	def	Selection
0.2	60.0	0.08	0.35	0.01	379.0	0.0	0.01	0.01	0.23	0.08	0.35	0.06	0.06	0.35	0.08	0.06	1	0.000000	False
0.2	60.0	0.08	0.4	0.01	379.0	0.0	0.01	0.01	0.23	0.08	0.304	0.06	0.06	0.35	0.08	0.06	2	0.000000	False
0.2	60.0	0.08	0.4	0.01	381.0	0.0	0.01	0.01	0.23	0.08	0.304	0.06	0.06	0.35	0.08	0.06	2	100.000000	False
0.2	60.0	0.08	0.4	0.01	381.0	0.0	0.01	0.01	0.23	0.08	0.304	0.12	0.12	0.35	0.08	0.12	8	37.500000	False
0.2	60.0	0.08	0.4	0.01	381.0	0.0	0.01	0.01	0.23	0.08	0.304	0.12	0.12	0.3	0.08	0.12	5	0.000000	False
0.2	60.0	0.07	0.4	0.01	381.0	0.0	0.01	0.01	0.23	0.07	0.304	0.12	0.12	0.3	0.07	0.12	20	30.000000	True
0.25	60.0	0.07	0.4	0.01	381.0	0.0	0.01	0.01	0.23	0.07	0.304	0.12	0.12	0.3	0.07	0.12	3	66.666667	False
0.25	62.0	0.07	0.4	0.01	381.0	0.0	0.01	0.01	0.23	0.07	0.304	0.12	0.12	0.3	0.07	0.12	1	0.000000	False
0.25	58.0	0.07	0.4	0.01	381.0	0.0	0.01	0.01	0.23	0.07	0.304	0.12	0.12	0.3	0.07	0.12	20	40.000000	True
0.28	58.0	0.07	0.4	0.01	381.0	0.0	0.01	0.01	0.25	0.07	0.304	0.12	0.12	0.3	0.07	0.12	3	0.000000	False
0.28	58.0	0.07	0.4	0.0	381.0	0.0	0.0	0.25	0.07	0.304	0.12	0.12	0.3	0.07	0.12	74	58.108108	True	
0.44	58.0	0.07	0.4	0.0	381.0	0.0	0.0	0.25	0.07	0.304	0.12	0.12	0.3	0.07	0.12	28	50.000000	True	

Fig. 2. Characterization of the partition A's clusters and identification of the problematic ones.

The last three columns represented in Figure 2 are, for each cluster in partition A, the defect rate, the number of tires in the cluster, and the characterization of a problematic cluster. These new attributes are computed in the following step of the methodology, the **Defect rate analysis**. On the same way, within each partition, clusters are created to group the products.

3) *Defect rate analysis*: For each partition, the defect rate of each cluster in the partition is calculated following equation 2. It corresponds to the ratio of defective tires to the number of tires in each cluster.

As an example, problematic clusters are defined by a defect rate greater than 20% and a minimum number of 30 products. With those values, in Figure 2, 4 clusters meet the minimum defect rate (in yellow) and the minimum number of products (in green) and are therefore designated as problematic (in blue) in partition A. A descriptive analysis of the root causes leading to these clusters of defects will be performed.

4) *Plausible root cause identification*: The problematic clusters identified are characterized according to the value of their production parameters present in the **Features** dataset. The CART decision tree algorithm is used to separate the problematic clusters from the other clusters present in the partition. The stopping criterion for branch separation is set at a minimum reduction in impurity $imp = 0.001$.

As illustrated in Figure 3, the nodes of the decision tree correspond to the production parameters of the tires.

The analysis of the path leading to problematic clusters in the tree enables the development of possible avenues of

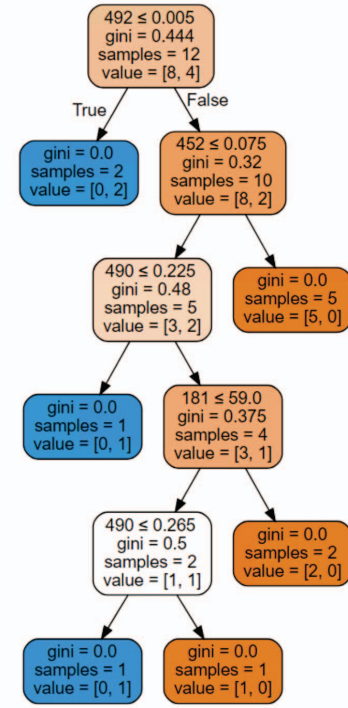


Fig. 3. Decision tree for partition A separating problematic/nonproblematic clusters based on tire production variable combinations.

reflection about the reasons for which a tire of partition A is in one of the problematic clusters previously identified. Indeed, the analysis of the tree proposed in Figure 3 shows that the problematic clusters of partition A (in blue) are associated with the following logical rules concerning the value of production parameters **181**, **452**, **490** et **492** :

- $492 \leq 0.005$
- $492 > 0.005$ AND $452 \leq 0.075$ AND $490 \leq 0.225$
- $492 > 0.005$ AND $452 \leq 0.075$ AND $181 \leq 59$ AND $0.225 < 490 \leq 0.265$

Information on the importance of a production parameter for production quality can also be gathered by calculating the frequency of appearance of a parameter in the tree. The more frequently the parameter is used to separate problematic clusters from others, the more critical the parameter can be considered to be for production quality.

In Figure 3, the parameter **490** is used twice in the tree. Thus, this parameter is more important than the others in identifying problematic clusters and explains their high defect rate. It can be deduced that this production parameter is a good candidate to explain the defects in partition A.

This analysis is repeated on all partitions with the highest defect rates in our dataset. The aim is to determine the parameters most frequently used to explain the high defect rate of problematic clusters. The overall frequency of occurrence of each parameter is illustrated in Figure 4, calculated by compiling the decision trees of all partitions together.

Parametre	Occurance parametre
0	489
1	473
2	491
3	490
4	476
5	487
6	475
7	488
8	474
9	464

Fig. 4. Importance of parameters for the seven tire references with the highest defect rate

It is interesting to note that the first nine parameters shown in figure 4 all relate to an operation performed on the product sidewall. Therefore, this operation appears to be the root cause of the defects in the assembly process. This observation was confirmed by expert knowledge represented by the industrial partner.

B. Root causes of defects in automobile production

Between January and April 2021, a total of 63 111 cars were assembled on a multi-product assembly line that exhibited high variability. Indeed, only three perfectly identical cars were registered during this period.

The studied section of the assembly line consists of 103 stations, divided into four sub-sections, allowing for the execution of 6 520 different operations. 11 265 defects in 959 categories were observed during the study period.

In this study, analysis partitions are based on the datasets used to cluster family products (**AVO**, **WS_load** and **WSS_load**) as well as the subsection of the assembly lines. These partitions are examined independently throughout the remainder of the case study.

1) *Data preparation*: First, the product specification data are prepared. For each assembled car, 8 attributes respectively representing the main characteristics of the vehicles, such as the engine reference, the car line, or its color, are binarized, generating a binary data set **Features** of size $[63111 \times 155]$. The attribute representing the operations performed on each car is also binarized, generating a binary data set **AVO** of size $[63111 \times 6520]$.

Then, scheduling data is used to generate a data set **WS_load** of size $[63111 \times 103]$ corresponding to the normalized workload (*Min/Max*) per workstation and car. Another data set **WSS_load** of size $[63111 \times 4]$, represents the workload per sub-section of the assembly line. It is generated by summing the previously calculated normalized workload for all stations belonging to the same section.

Finally, in the quality data, 11 265 defects divided into 959 categories are identified. Defects relating to cars assembled

outside normal production runs, based on the company knowledge, are not considered. As a result, 7 083 defects in 577 categories are recorded. In this case study, a car is considered to be defective if it has at least one defect, which represents 9% of the cars present in the studied database. The binarized data set **Quality** of size $[63111 \times 1]$ is used to determine whether the assembled car is defective or not.

In this case study, no data regarding the assembly process is considered.

2) *Product family identification*: Since all cars are unique, and no internal company knowledge provides a classification system, AHC clustering is employed to cluster cars based on the similarity of workload (**WS_load** and **WSS_load**) and operations (**AVO**).

Table I summarizes the different partitionings used to group vehicles. Workload-based partitions include partitions 1 to 6, while operation-based partitions include partitions 7 to 11. Clusters in partitions 1 to 5 are generated using the **WS_load** dataset to group vehicles according to the workload at each station. In partition 6, clusters are generated using the **WSS_load** dataset to group vehicles according to the cumulative workload for each sub-section of the assembly line. Finally, partitions 7 to 11 are based on the **AVO** dataset to group vehicles according to the similarity of operations performed on each of them.

TABLE I
PARTITIONING OVERVIEW BASED ON WORKSTATION LOAD AND OPERATION DATA

Partition	Description	Data source	Nb clusters
1	All data from WS_load	WS_load	2
2	Data from workstations in the first sub-section only	WS_load	2
3	Data from workstations in the second sub-section only	WS_load	2
4	Data from workstations in the third sub-section only	WS_load	4
5	Data from workstations in the fourth sub-section only	WS_load	2
6	All data from WSS_load	WSS_load	5
7	All data from AVO	AVO	4
8	Data from operations in the first sub-section only	AVO	4
9	Data from operations in the second sub-section only	AVO	2
10	Data from operations in the third sub-section only	AVO	4
11	Data from operation in the fourth sub-section only	AVO	2

The number of clusters associated with each partition is determined using the silhouette method as illustrated for partition 4 in Figure 5.

Partition 4 will be used to illustrate the results obtained in the rest of the case study.

After using the AHC algorithm to create the previously determined number of clusters for each of the 11 data partitions,

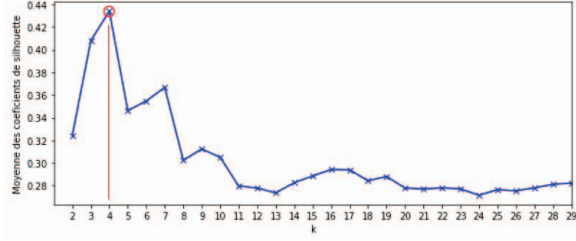


Fig. 5. Silhouette method for partitioning 4 based on sub-section 3 standardized workloads

a statistical analysis is performed on each partition using the **Quality** dataset for the defect rate analysis.

3) *Defect rate analysis*: First, the defect rate of each cluster in the partition is calculated following equation 2. It corresponds to the ratio of defective cars to the number of cars in each cluster.

Then, the defect rate is calculated for each transition between clusters in the partition, following equation 3.

The results of the statistical analysis are presented in the form of cluster defect rate density maps. They can then be compared with the average defect rate present in the data set (9%) to identify problematic clusters as shown in figure 6.

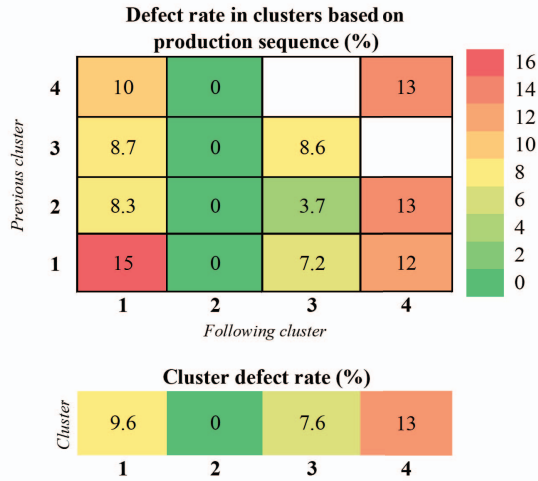


Fig. 6. Pair of density maps for partitioning 4 : standardized workload in subsection 3.

Figure 6 shows that Cluster 4 has a defect rate of $Defect_4 = 13\%$, which is four points higher than the defect rate in the dataset. Furthermore, it can be noted that the consecutive production of two cars from Cluster 1 results in a defect rate of $Defect_{1 \rightarrow 1} = 15\%$, which is 5% higher than the defect rate of Cluster 1 ($Defect_1 = 9.6\%$). According to these observations, in this partitioning, the clusters 1 and 4 seem to be problematic.

Cluster 2 has a failure rate of $Defect_2 = 0\%$. This result is consistent with the previous data preparation step, as Group 2 exclusively contains cars assembled outside normal

production, for which defects have been excluded in this study (see data preparation step).

4) *Plausible root cause identification*: The problematic clusters identified are finally characterized according to the characteristics of the cars present in the dataset **Features**. The CART decision tree algorithm is used to separate the problematic clusters from the other clusters present in the partition. The stopping criterion for branch separation is set at a minimum reduction in impurity $imp = 0.001$.

As illustrated in Figure 7, the nodes of the decision tree correspond to the principal characteristics of the cars.

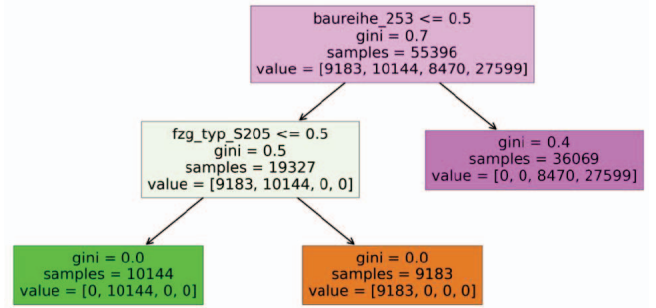


Fig. 7. Classification of the different clusters in partition 4 (standardized workload in subsection 3) according to car characteristics.

The analysis of the path leading to problematic clusters in the tree enables the development of possible avenues of reflection about the reasons why a car in partition 4 is in one of the problematic clusters previously identified. According to Figure 7, Cluster 1 (in orange) contains cars that do not possess the feature *Baureihe_235*, but do have the feature *fzg_typ_S205*. Cluster 4 (in purple), contains cars with the *Baureihe_235* feature.

In light of these observations regarding Cluster 4, further consideration should be given to the effect of *Baureihe_235* on the defect rate of assembled vehicles. Vehicles with this characteristic have a higher defect rate than the general production average. Thus, this trait could be a root cause of defects.

Finally, a focus on the production sequence provides a better understanding of the possible causes of defects in Group 1. Indeed, one can see that the consecutive production of two cars without the characteristic *Baureihe_235*, but with the characteristic *Fzg_typ_S205* (cars from Cluster 1), leads to an increase in the defect rate of 4%. In this case, the cause of the defects is more likely to be related to the production sequence.

V. DISCUSSION

The proposed methodology was applied to two real-world case studies to validate its applicability. In both case studies, a decision tree was used to identify the root causes of defects that resulted in clusters of high defect rates. Industry experts validated the identified root causes of defects, demonstrating the validity of the generated rules.

However, the proposed methodology identifies root causes from the data, but their validity is not measurable. Therefore, expertise in root cause analysis, as emphasized by Ito et al. [30], remains challenging and still necessitates expert assessments. Moreover, because root cause analysis is usually performed in the context of an occurring problem, it is also necessary for the methodology to be used to be able to identify a data sample representing the context in which the problem occurred to be used as input to the methodology. The size of the sample and the time frame of the study significantly impact the analyses performed. For example, a new production strategy or the removal of a machine could have a huge impact on the model's robustness. Therefore, formalizing the criteria to assess the pertinence of the chosen data should be studied further to develop a relevant analysis framework for future continuous improvement projects.

This importance in the choice of data used to identify the root causes of defects leads to a final limitation of the proposed method, which lies in the need to regenerate the tree for each new analysis. When studying a new data partition (depending on the product reference or family), the **Defect rate analysis** and **Plausible root cause identification** steps must be repeated.

Conducting multiple analyses can be time-consuming and resource-intensive. However, it improves the traditional RCA method regarding time consumption and identification of possible leads (cf. section II). Although the use of a generalizable pipeline of machine learning (ML) models was tempting, especially to be able to validate using classical metrics (RMSE, MAPE, etc.), it is not the aim of this article. The implementation of a predictive or classification model of an apparent problem would be aimed at avoiding the occurrence of the it in the first place, which is not the context in which RCA is used. The fact that RCA is only performed under circumstances that deviate from the standard, supports our descriptive method as an avenue of choice.

VI. CONCLUSION

The need for mass customization and cost reduction is always relevant to meet the demands of increasingly demanding consumers. Multi-product assembly lines enable the production of a wide variety of products, but this strength is also a weakness since it generates a large number of defects, thus reducing cost efficiency.

In contrast to conventional defect root cause analysis methods, which struggle to deliver satisfactory results on such assembly lines, this research proposes a four-step methodology for data preparation and descriptive analysis of defect root causes, specifically applied to multi-product assembly lines.

Production data from the company's information systems is first prepared and segmented using either AHC algorithms or business knowledge. Then, the defect rate for each product group and the defect rate associated with the transition between two product sequences during production are calculated. Finally, a CART decision tree allows for descriptive analysis of the root causes of defects by elaborating logical rules linked

to the product or process characteristics that lead to defective product families.

Future studies considering the use of other explainable machine learning algorithms, such as Bayesian networks, for classification should be undertaken. A comparison of the results and limitations proposed by each of these algorithms would be relevant to developing tools for automatic analysis of root causes of defects in the context of multi-product assembly lines. Moreover, the utilization of more opaque algorithms, such as neural networks, in conjunction with ad-hoc explicability models, such as SHAP analysis, could prove beneficial in enhancing classification performance while ensuring the explicability necessary for the specific problem at hand.

The methodology was successfully applied to real production data in two case studies, enabling the detection of root causes of defects. These causes were then validated by the partner company. The combination of multiple data enhancement techniques proposed in the methodology can improve the process of identifying the root causes of defects on a multi-product assembly line, even if the number of defects per reference is very low.

However, the proposed methodology still has limitations that need to be addressed. These include the need for experts to assess the validity of the root causes of defects, the dependence of results on the data used (including sample size and time frame), and the need to regenerate the tree for each new analysis.

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